

Bharatiya Antariksh Hackathon 2026

Team Name: _____

Team Leader Name: _____

Problem Statement: Antigravity Track - Autonomous Satellite Intelligence System

Team Members

Team Leader:

Name: _____

College: _____

Team Member-1:

Name: _____

College: _____

Team Member-2:

Name: _____

College: _____

Team Member-3:

Name: _____

College: _____

Opportunity

How different is it from any of the other existing ideas?

Traditional ground operations rely on human operators manually reading 2D telemetry dashboards and consulting manuals. VyomOS shifts this paradigm by moving intelligence to the edge, processing anomalies autonomously using a local LLM, and providing mitigating commands instantly without internet dependency.

How will it be able to solve the problem?

It reduces human cognitive load and reaction time from minutes to milliseconds. By automating detection, explanation, and mitigation, VyomOS prevents catastrophic failures during high-stress events like solar flares or communication blackouts.

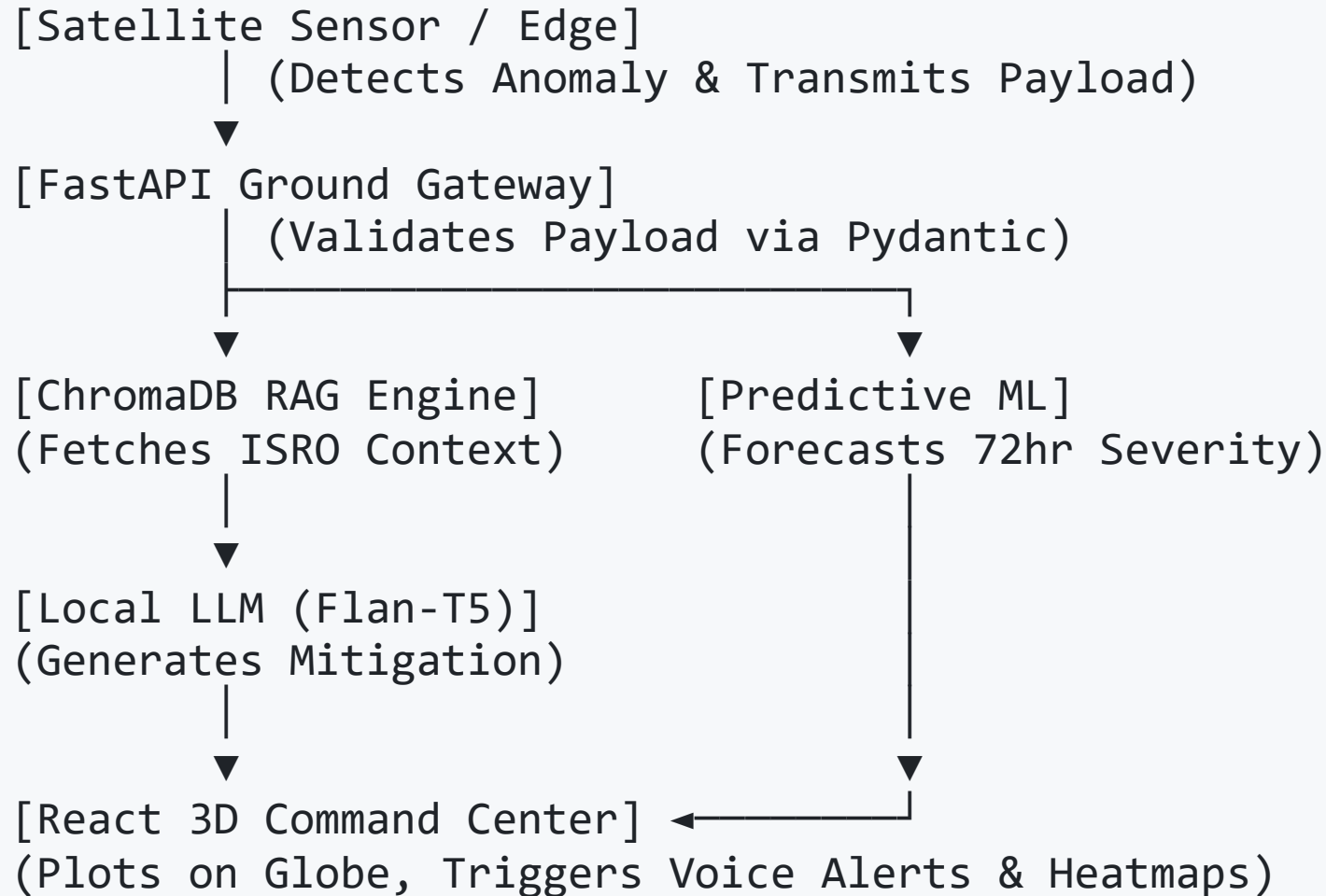
USP of the proposed solution:

- 100% Data Sovereignty (Zero reliance on OpenAI or external clouds)
- Multi-Satellite Cascading Anomaly Correlation

List of features offered by the solution

- **3D Digital Twin:** Real-time WebGL visualization of ISRO's fleet using live Celestrak TLEs.
- **Autonomous AI Mitigation:** On-device LLM (Flan-T5) generates instant recovery strategies.
- **ISRO RAG Engine:** Mitigation is grounded in actual ISRO mission protocols using ChromaDB.
- **Predictive Forecasting:** ML model predicts anomaly severity 72 hours into the future.
- **NASA DONKI Integration:** Automatically triggers alerts based on real-world space weather.
- **Telemetry Gauges & Heatmaps:** Live tracking of battery, solar output, and historical anomaly density.

Process flow diagram / Use-case diagram



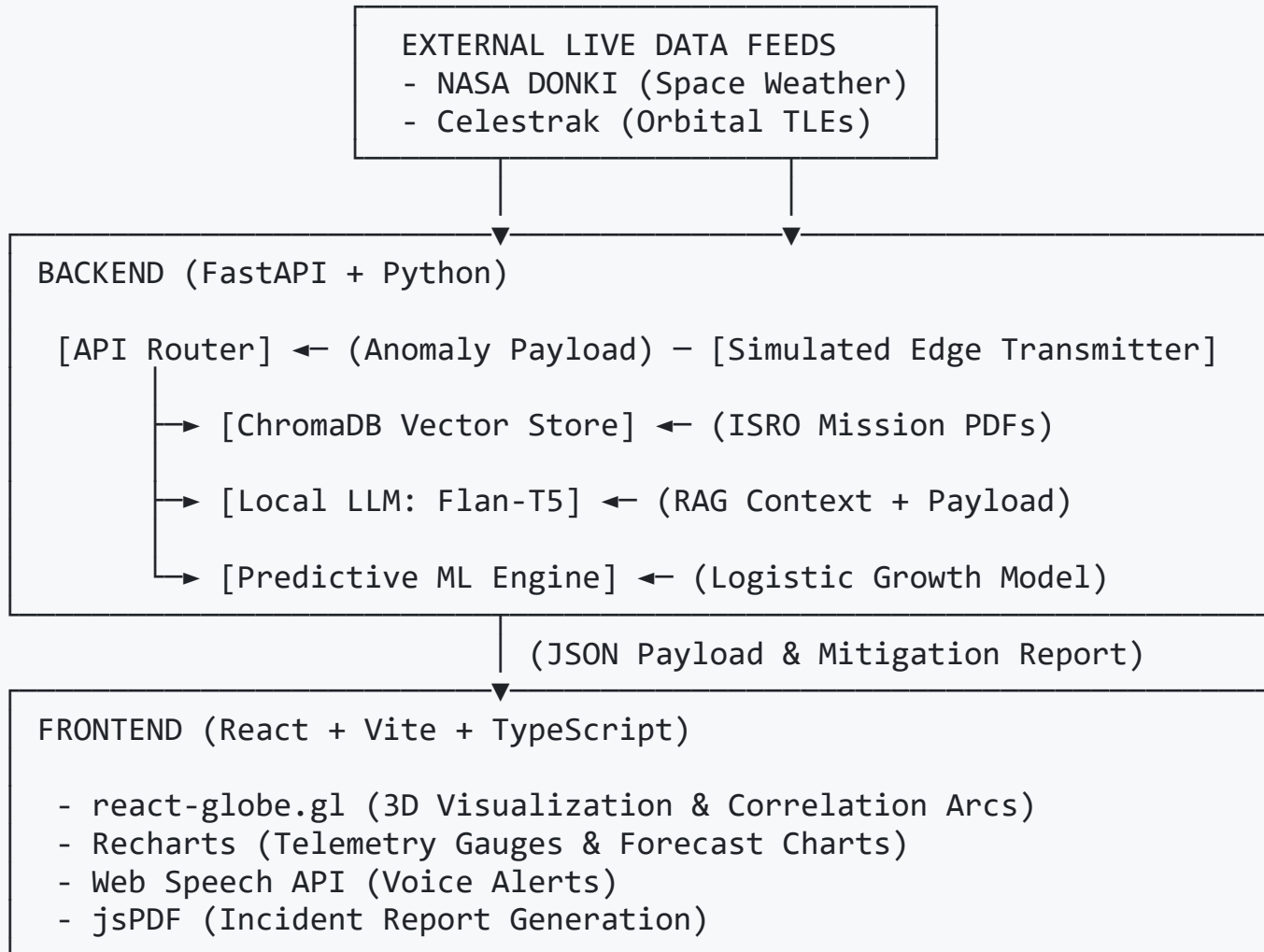
Wireframes / Mock diagrams of the proposed solution

Command Center Layout:

1. **Top Bar:** Ticker tape showing live X-Ray Flux and Flare Classes, along with an overarching Alert Banner (Normal/Watch/Warning/Emergency).
2. **Left Panel (Analytics):** Satellite status matrix, NavIC degradation impact, and live Telemetry Gauges (Battery, Solar, Temp) rendered via Recharts.
3. **Center/Right (Digital Twin):** A massive interactive 3D Globe rendering ISRO satellites, orbital paths, Day/Night terminators, and pulsing Red Alert shockwaves during anomalies.
4. **Bottom Panel:** 72-hour predictive anomaly severity chart using a logistic growth model.

5. **Footer:** A minimalist footer containing the project name, version (v1.0.0), and a "Data as of: 2025-01-16 14:30 UTC" timestamp.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Frontend:

- React 18, TypeScript, Vite
- Tailwind CSS (Glassmorphic UI)
- `react-globe.gl` & `three.js` (3D rendering)
- `recharts` & `d3` (Data visualization)
- `satellite.js` (Orbital mechanics)

Backend:

- Python 3.10+, FastAPI, Uvicorn
- Pydantic (Data validation)

AI & Machine Learning:

Estimated implementation cost (optional)

Development & Software Cost:

- \$0 (VyomOS is built entirely on open-source frameworks, free public APIs like NASA DONKI, and locally hosted AI models).

Deployment Hosting (Command Center):

- \$0 for Hackathon (Render/Vercel Free Tiers).
- ~\$50/month (Production-grade cloud backend for ground station).

Edge Hardware (Per Satellite/Node):

- ~\$150 - \$300 (NVIDIA Jetson Nano or equivalent space-hardened edge compute module for running the local LLM and anomaly detection on-orbit).

Total Prototype Cost: \$0